Analisis data

Data berisi informasi dari segala sesuatu informasi mengenai tentang study diluarnegri. Informasi berisi:

1. Negara,Kota
2. Universitas, Jurusan, tingkatan
3. Biaya kesaharian, Biaya kuliah, Biaya Rumah ect

Pemanggilan library

In [1]:
```
import pandas as pd
import numpy as np
```

In [2]:
```
df=pd.read_csv("International_Education_Costs.csv")
df.shape
```

Out[2]: (907, 12)

In [3]:
```
df.head()
```

Out[3]:

	Country	City	University	Program	Level	Duration_Years	Tuition_USD
0	USA	Cambridge	Harvard University	Computer Science	Master	2.0	55400
1	UK	London	Imperial College London	Data Science	Master	1.0	41200
2	Canada	Toronto	University of Toronto	Business Analytics	Master	2.0	38500
3	Australia	Melbourne	University of Melbourne	Engineering	Master	2.0	42000
4	Germany	Munich	Technical University of Munich	Mechanical Engineering	Master	2.0	500

In [4]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 907 entries, 0 to 906
Data columns (total 12 columns):
 #   Column             Non-Null Count  Dtype  
---  ------             --------------  -----  
 0   Country            907 non-null    object 
 1   City               907 non-null    object 
 2   University         907 non-null    object 
 3   Program            907 non-null    object 
 4   Level              907 non-null    object 
 5   Duration_Years     907 non-null    float64
 6   Tuition_USD        907 non-null    int64  
 7   Living_Cost_Index  907 non-null    float64
 8   Rent_USD           907 non-null    int64  
 9   Visa_Fee_USD       907 non-null    int64  
 10  Insurance_USD      907 non-null    int64  
 11  Exchange_Rate      907 non-null    float64
dtypes: float64(3), int64(4), object(5)
memory usage: 85.2+ KB
```

data memiliki 12 column dan 907 baris

Cleaning data set

1. cek dan ganti nilai null

In [5]:
```
df.isnull().sum()
```

Out[5]:
```
Country              0
City                 0
University           0
Program              0
Level                0
Duration_Years       0
Tuition_USD          0
Living_Cost_Index    0
Rent_USD             0
Visa_Fee_USD         0
Insurance_USD        0
Exchange_Rate        0
dtype: int64
```

2. cek dan hapus data duplicate

In [6]:
```
df.duplicated().sum()
```

Out[6]:
```
np.int64(0)
```

Analisis data

1. penambahan variable columns untuk menjelaskan harga sewa selama 1 tahun

In [7]:
```
df["Rent/year"]= df["Rent_USD"]* 12
df["total"]=df["Tuition_USD"]+ df["Rent/year"]+df["Visa_Fee_USD"]
df.head()
```

Out[7]:

	Country	City	University	Program	Level	Duration_Years	Tuition_USD
0	USA	Cambridge	Harvard University	Computer Science	Master	2.0	55400
1	UK	London	Imperial College London	Data Science	Master	1.0	41200
2	Canada	Toronto	University of Toronto	Business Analytics	Master	2.0	38500
3	Australia	Melbourne	University of Melbourne	Engineering	Master	2.0	42000
4	Germany	Munich	Technical University of Munich	Mechanical Engineering	Master	2.0	500

2. Central Tedency

In [8]:
```
df.describe()
```

Out[8]:

	Duration_Years	Tuition_USD	Living_Cost_Index	Rent_USD	Visa_Fee_USD	I
count	907.000000	907.000000	907.000000	907.000000	907.000000	
mean	2.836825	16705.016538	64.437486	969.206174	211.396913	
std	0.945449	16582.385275	14.056333	517.154752	143.435740	
min	1.000000	0.000000	27.800000	150.000000	40.000000	
25%	2.000000	2850.000000	56.300000	545.000000	100.000000	
50%	3.000000	7500.000000	67.500000	900.000000	160.000000	
75%	4.000000	31100.000000	72.200000	1300.000000	240.000000	
max	5.000000	62000.000000	122.400000	2500.000000	490.000000	

In [9]:
```
# mean costan
analist_1=df.describe()

analist_1
```

Out[9]:

	Duration_Years	Tuition_USD	Living_Cost_Index	Rent_USD	Visa_Fee_USD	I
count	907.000000	907.000000	907.000000	907.000000	907.000000	
mean	2.836825	16705.016538	64.437486	969.206174	211.396913	
std	0.945449	16582.385275	14.056333	517.154752	143.435740	
min	1.000000	0.000000	27.800000	150.000000	40.000000	
25%	2.000000	2850.000000	56.300000	545.000000	100.000000	
50%	3.000000	7500.000000	67.500000	900.000000	160.000000	
75%	4.000000	31100.000000	72.200000	1300.000000	240.000000	
max	5.000000	62000.000000	122.400000	2500.000000	490.000000	

In [10]:

```
analist =pd.DataFrame(
    {
        "variable":["Tuition_USD", "Living_Cost_Index", "Exchange_Rate","
        "Means": [analist_1["Tuition_USD"].loc["mean"],
                  analist_1["Living_Cost_Index"].loc["mean"],
                  analist_1["Exchange_Rate"].loc["mean"],
                  analist_1["total"].loc["mean"]]
}
)
#Table info Average Cost in Students Interantional
print ("Average for Cost International Students")

analist.head()
```

Average for Cost International Students

Out[10]:

	variable	Means
0	Tuition_USD	16705.016538
1	Living_Cost_Index	64.437486
2	Exchange_Rate	623.000695
3	Total/year	28546.887541

3. Top negara yang menyediakan pendidikan international

In [11]:

```
#Countries
from collections  import Counter
country= Counter (df["Country"])
x=[name for name in country.keys()]
y =[country[name]for name in x]
country=pd.DataFrame(
    {
        "Country":x,
        "value":y
    }
)
country.to_csv("country.csv")
top_ =country.sort_values (by ='value', ascending= False)
top_.head(10)
```

Out[11]:

	Country	value
1	UK	93
3	Australia	86
0	USA	78
2	Canada	76
4	Germany	33
8	France	27
13	South Korea	23
6	Netherlands	21
9	Switzerland	20
7	Singapore	18

3. Informasi mengenai tingkatan pendidikan yang tersedia

In [12]:

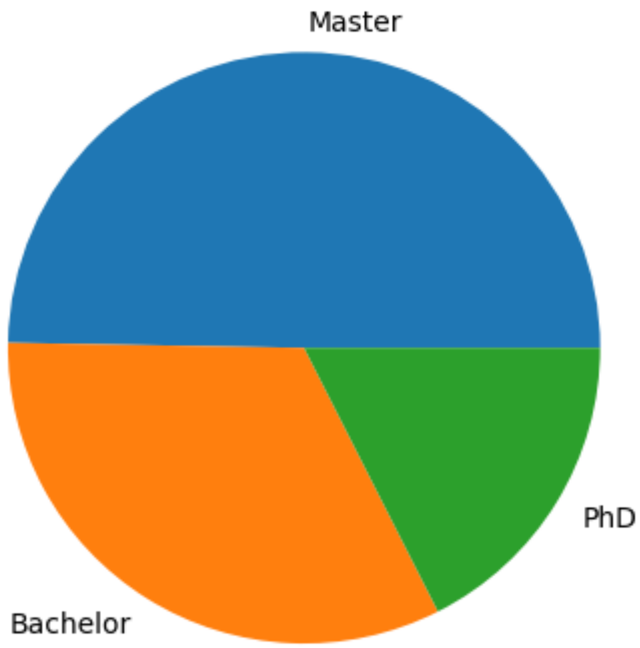
```
# information degrees
import matplotlib.pyplot as plt
# phd/bachelor/ Majister
Data3=Counter(df["Level"])
x=[level for level in Data3.keys()]
y=[Data3[name] for name in x]
data3=pd.DataFrame({
    "level":x,
    "quantity":y
})
data3.to_csv("level1", index=False)
print("level and quantity")
plt.pie(y,labels=x)
plt.show()

#mean duration study/level

data4= df.groupby(["Level"])["Duration_Years"].mean()

print (data4)
data4.to_csv("duration_level.csv")
```

```
level and quantity
```




```
Level
Bachelor    3.464646
Master      1.984479
PhD         4.081761
Name: Duration_Years, dtype: float64
```

In [13]:

```
df.head()
```

Out[13]:

	Country	City	University	Program	Level	Duration_Years	Tuition_USD
0	USA	Cambridge	Harvard University	Computer Science	Master	2.0	55400
1	UK	London	Imperial College London	Data Science	Master	1.0	41200
2	Canada	Toronto	University of Toronto	Business Analytics	Master	2.0	38500
3	Australia	Melbourne	University of Melbourne	Engineering	Master	2.0	42000
4	Germany	Munich	Technical University of Munich	Mechanical Engineering	Master	2.0	500

4. Universitas gratis

In [14]:

```
# information univesty tuition==0
data5 =df.loc[df["Tuition_USD"]==0, ["University","Program","Level","Coun
data5
```

Out[14]:

	University	Program	Level	Country	total
10	KTH Royal Institute	Sustainable Technology	Master	Sweden	14510
11	University of Copenhagen	Bioinformatics	Master	Denmark	15720
39	Lund University	Neuroscience	PhD	Sweden	12110
45	Aalborg University	Computer Science	Bachelor	Denmark	10920
54	Free University of Berlin	International Business	Bachelor	Germany	10875
...	...	...	...	...	...
754	National University of La Plata	Software Development	Bachelor	Argentina	2970
761	National University of Cuyo	Data Analytics	PhD	Argentina	2850
770	Universidad de la Republica	Computer Engineering	PhD	Uruguay	5490
794	UTEC	Information Systems	Bachelor	Uruguay	4290
802	Universidad de Paysandu	Software Engineering	Master	Uruguay	4050

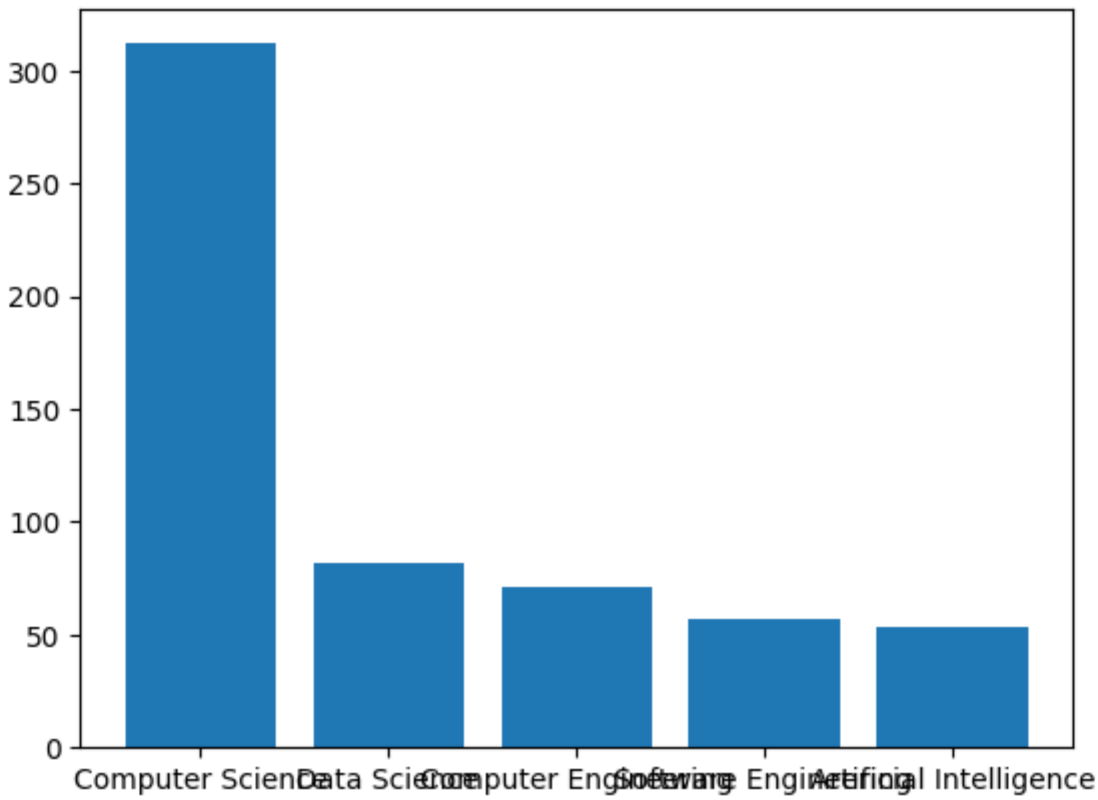
103 rows × 5 columns

5. prodi yang diminati

In [15]:

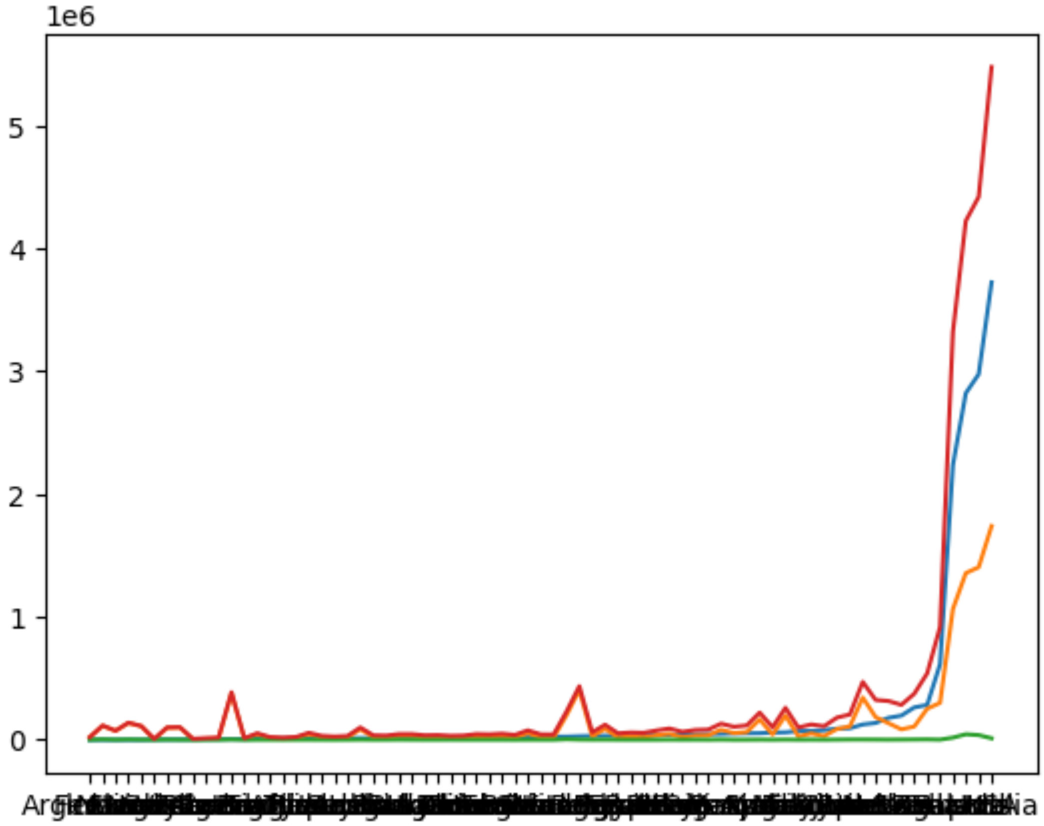
```
# most
Program=df["Program"]
Program=Counter(Program)
x=[k for k in Program.keys()]
y=[Program[k] for k in x ]
data6= pd.DataFrame(
    {
        "Program Study":x,
        "Quantity":y
    }
)
data6 = data6.sort_values ("Quantity", ascending=True)

data6_1=data6.sort_values("Quantity",ascending=False).head(5)
plt.bar(data6_1["Program Study"],data6_1["Quantity"])
plt.show()
```



6. Kebutuhan semua aspek keuagan pendidikan Internationl

In [20]:
```
data7=df.groupby(["Country"])[["Tuition_USD","Rent/year","Visa_Fee_USD","
data7=pd.DataFrame(data7)
data7=data7.sort_values("Tuition_USD", ascending=True)
data7.to_csv("data7.csv")
data7_ =pd.read_csv("data7.csv").sort_values("Tuition_USD")
plt.plot(data7_["Country"],data7_["Tuition_USD"])
plt.plot(data7_["Country"],data7_["Rent/year"])
plt.plot(data7_["Country"],data7_["Visa_Fee_USD"])
plt.plot(data7_["Country"],data7_["total"])
plt.show ()
```



Grafik diurutkan berdasarkan biaya kuliah dari yang terkecil ke yang terbesar dikelompokkan dalam satu negara. Semua aspek biaya kehidupan meningkat jika biaya kuliahnya besar. Kebutuhan yang besar juga dipengaruh oleh wilayah juga.